

CHEMISTRY

(EEE, ECE, CSE)

(Model Paper-II)**Time: 3 hours****Max Marks: 70M**

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART - A			10 X 2 = 20 M		
			CO	BTL	Marks
I.	a)	Write the significance of Ψ and Ψ^2 .	CO1	L2	2 M
	b)	Calculate the bond order in NO molecule.	CO1	L2	2 M
	c)	What is an elemental semiconductor? Give examples.	CO2	L1	2 M
	d)	Write any four applications of coreshell nanoparticles.	CO2	L2	2 M
	e)	List the advantages of potentiometric titrations.	CO3	L4	2 M
	f)	Write any four uses of zinc-air battery.	CO3	L2	2 M
	g)	Write any four differences between thermoplastics and thermosetting plastics.	CO4	L2	2 M
	h)	Identify the monomers present in the following polymers (a) Buna-S (b) Polyglycolic acid	CO4	L3	2 M
	i)	What is an electromagnetic spectrum? Identify the wavelengths of its various regions.	CO5	L3	2 M
	j)	What is Chromatography? Write its principle.	CO5	L2	2 M
PART - B			5 X 10 = 50 M		
UNIT - I					
1.	a)	Solve the Schrodinger wave equation for a particle in a one-dimensional box with zero potential energy inside the box.	CO1	L3	5 M
	b)	Describe the energy level diagram of CO. Identify its magnetic nature.	CO1	L3	5 M
(OR)					
2.	a)	Discuss the experimental verification of the dual nature of electrons.	CO1	L2	5 M
	b)	Describe the π -molecular orbitals of Benzene. Identify its HOMO and LUMO.	CO1	L3	5 M
UNIT - II					
3.	a)	What is a supercapacitor? Explain about the Hybride capacitor.	CO2	L2	5 M
	b)	Outline the properties and applications of Graphene.	CO2	L2	5 M
(OR)					
4.	a)	Write short notes on EDLC. List the applications of supercapacitors.	CO2	L3	5 M
	b)	What is Hierarchical Porous Carbon(HPC)? List its applications.	CO2	L3	5 M
UNIT - III					
5.	a)	Write the Nernst equation. calculate the EMF of the following cell at 298 K. $\text{Zn(s)} / \text{Zn}^{2+} (0.01 \text{ M}) // \text{Pb}^{2+} (0.001 \text{ M}) / \text{Pb(s)}$ If $E^{\circ}_{\text{Zn}^{2+} / \text{Zn}} = -0.76\text{V}$; $E^{\circ}_{\text{Pb}^{+2} / \text{Pb}} = -0.13\text{V}$.	CO3	L3	5 M
	b)	What is a secondary cell? Discuss the construction, working and uses of lithium ion battery.	CO3	L2	5 M
(OR)					
6.	a)	Discuss theconductometric titrations in the following cases. a. Strong acid and strong base b. Weak acid and strong base	CO3	L2	5 M
	b)	What is a fuel cell? Explain the construction, working and	CO3	L2	5 M

		applications of polymer electrolyte membrane fuel cell.			
UNIT – IV					
7.	a)	Discuss the preparation and properties of Teflon. List its applications.	CO4	L4	5 M
	b)	What is a conducting polymer? Explain the mechanism of conduction in polyacetylene.	CO4	L2	5 M
(OR)					
8.	a)	Describe the mechanism of cationic polymerization with an example.	CO4	L2	5 M
	b)	Outline the preparation, properties, and applications of Buna-N.	CO4	L2	5 M
UNIT – V					
9.	a)	Describe the instrumentation of UV-Visible spectroscopy with a neat sketch.	CO5	L2	5 M
	b)	Illustrate the fundamental modes in IR spectroscopy.	CO5	L2	5 M
(OR)					
10.	a)	Why UV spectroscopy is called electronic spectroscopy? Explain briefly the electronic transitions in UV spectroscopy.	CO5	L2	5 M
	b)	Outline the applications of IR spectroscopy as an analytical tool.	CO5	L2	5 M

***** End of the Question Paper *****